

It's chunking! – And how it shapes the grammars of natural languages

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Abstract

Chunking is a domain-general information processing strategy in which the brain organizes and groups individual pieces of information into larger, manageable units or "chunks." This technique reduces the cognitive load and enables complex information to be processed quickly. In grammar theory, chunking is the relevant background issue for understanding and explaining invariant or type-defining structural properties. Structure dependence and recursion are direct reflections of the typical human chunking capacity. Efficient chunking is a selection criterion in the cognitive evolution of grammars. Universal right-branching, fillers preceding gaps, V-clustering, the Left-Left Constraint, and the restriction on the number of argument slots are reflections of chunking properties. Chunking is what creates the link between the study of grammatical structures and the studies of (neuro-)cognitive processes of information processing.

1. Background

Two ideas emerged side by side around 70 years ago without being linked together afterwards, although they deserved to be. One is the idea of *chunking* as a fundamental component of human information processing. The other idea is the *structure-dependence* of grammatical regularities, with *recursion* of grammatical rules and structures. Until today, Generative grammar claims recursion and structure dependence as its own domain-specific terrain since these two things are understood by Generative Grammar as genuinely linguistic and as reflections of linguistic universals conditioned by an innate UG. However, if the advocates of UG were to lean over the boundary wall of their self-segregated district, they would find that the neighbouring cognitive scientists know better.

For more than fifty years – see Chomsky (1972) – structure-dependence has been emphasized mantra-like as a universal of the human language faculty, the modelling of which would not be possible without UG: “*The only plausible conclusion, then is that structure-dependence is an innate property of the language faculty.*” Chomsky (2017: 5). The “only plausible conclusion” is derived from a comparison with an implausible alternative, namely *linearisation-based*, as a simpler(!) option. Linear *structures* are certainly simpler objects, but *rules* based on them are at least as complicated as structure-based rules.¹ Why is there no language that uses the conceptually simple operation of list inversion – a linearisation-based operation – for coding sentence mood for instance? Compared to aux inversion, everyone would immediately understand what is meant by re-ordering a sentence word by word from back to front. But we are unable to carry this out spontaneously and this is the relevant point. Our spontaneously available processing resources do not support list inversion.

As for recursion, “*we hypothesize that FLN [faculty of language in the narrow sense]_{HH} only includes recursion and is the only uniquely human component of the faculty of language*”. (Hauser

¹ Chomsky's favourite example is Aux-inversion. What would be the linearisation-based rule for a yes/no-question version of “*What happens once, happens again*”? First come, first served – as Chomsky suggests? This would sabotage the concomitant semantics where the fronted auxiliary belongs to the questioned predicate, not to mention the extraction constraints needed for narrowing the search space in filler-gap configurations.

et. al. 2002:1569). “*We suggest that FLN – the computational mechanism of recursion – is recently evolved and unique to our species*” (Hauser et. al. 2002:1573).

That was too bold a hypothesis to hold for long.² Biologists object: “*There is more empirical evidence of recursion in animal communication.*” (Barón Birchenall 2016: 22). In the meantime, experimental evidence is available showing that other species master tasks that involve recursion, too: “*We reveal that crows have recursive capacities; they perform on par with children and even outperform macaques. [...] These results demonstrate that recursive capabilities are not limited to the primate genealogy and may have occurred separately from or before human symbolic competence in different animal taxa.*” Liao et al. (2022: 1). A direct comparison between humans and monkeys (Ferrigno et al. 2020: 1) shows that “*the capacity itself is not unique to humans*”.

It is truly amazing that the fact that structure-dependence and recursion are intertwined seems to have received little attention so far. In fact, these are the two sides of the same coin, and the name of the cognitive currency is “chunking.” Incidentally, George Miller, then at Harvard, published his ground-breaking paper³ on chunking at the time when “Syntactic Structures” went to press.⁴

“The span of absolute judgment and the span of immediate memory impose severe limitations on the amount of information that we are able to receive, process, and remember. By organizing the stimulus input simultaneously into several dimensions and successively into a sequence of chunks, we manage to break (or at least stretch) this informational bottleneck.” (Miller 1956: 94).

Christiansen & Chater (2016, sect. 7) call it the *Now-or-Never Bottleneck*: “*Sequential information, at many levels of analysis, must rapidly be recoded to avoid being interfered with or overwritten by the deluge of subsequent material. To cope with the Now-or-Never bottleneck, the language system chunks new material as rapidly as possible at a range of increasingly abstract levels of representation.*”

Gobet et al. (2016: 1) present a basic taxonomy of the meanings of “chunk” and “chunking” across various fields of applications, ranging from cognitive science to computer science. “*A chunk typically refers to a single unit built from several smaller elements, and chunking to the process of creating a chunk,*” that is, “*a collection of elements having strong associations with one another, but weak associations with elements within other chunks.*” What we are concerned with here is “automatic chunking” (in contrast to deliberate chunking, as for instance in mnemotechnics). Automatic chunking is unconscious, implicit, and continuous. Chunking is essential for the efficient on-line processing of any kind of information, including linguistic input, as it efficiently reduces the cognitive load (Norris & Kalm 2021).

² Note, that Chomsky contradicts his own claim: If “*structure-dependence is an innate property of the language faculty,*” then recursion cannot be “*the only uniquely human component of the faculty of language*”. The two concepts are logically independent.

³ 43 600 citations according to PoP, based on Google Scholar (by Nov. 1, 2024)

⁴ At this time, Miller and Chomsky spent a summer together at Stanford University, their two families living in a fraternity house. Chomsky, Miller said, is a “*brilliant man, [but] he likes to argue, and I don't.*” He added, Chomsky “*turned me into a linguist.*” Miller did not turn him into a psychologist. [Richard Hébert 2006]

Recursion is a predictable concomitant effect of chunking. Structure recursion is an inevitable outcome whenever a complex chunk is part of a chunk of the same category. A clause inside a clause is recursion, but also a triangle consisting of triangles that consist of triangles we easily perceive as a recursive structure. And if I think what you as a reader may think that I am thinking when writing this, it is recursion. The ability to master recursion is not domain specific. It is a domain general capacity of humans and – to a certain degree – also of some animals. It presupposes the neuro-cognitive ability to build up *nested* mental representations.

Perceptual chunking in language processing is automatic chunking. Basically, a recoding operation is operative when (human) brains are engaged in processing input sequences (of any modality). Small(er) elements are grouped into larger, hierarchically organized units, i.e. chunks, and in still larger chunks (super-chunks) in a partially recursive fashion. Conversely, complex units, perceived as complex chunks, are divided up into less complex units. The chunks in which the information is grouped are an effective way of increasing the capacity of working memory capacities, enhancing the efficiency at the input bottleneck.

First-order chunking operates on the terminal-item level. It involves grouping small units into larger units such as syllables (in phonology) or phrases (in syntax). As sketched in Figure 1 from Rabinovich et al. (2014: 3), this basic level of chunking relates the “*elementary items network*” to the “*chunking network*”. Second-order chunking refers to organizing larger chunks into even more complex structures. It relates the “*chunking network*” to the “*superchunk network*,” creating hierarchical relationships between chunks. Together, both levels of chunking mirror the hierarchical structures that become manifest in natural language grammars with their box-in-box-structured phrasal architectures.

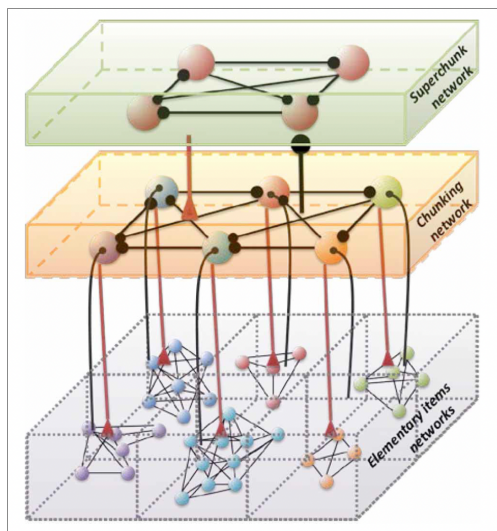


Figure 1: **The ECS-model of chunking**
(Rabinovich et al. 2014: 3)

“Architecture of the three-level cognitive network responsible for the grouping of informational items. [...]”

Black circles represent inhibitory connections; triangles represent excitatory connections responsible for the choosing of the informational items. Spheres represent the informational items or units (metastable states).

Different colors indicate different chunks. All connections inside the elementary items are inhibitory.”

Note that chunking is a distributed activity in our brain that consists of many similar, simultaneous information processing activities. Crucially, chunking is an essential feature of language processing, in perception, production, and acquisition. It works at the phonetic-phonology interface (segmental level) as well as in phonological structuring from syllables to phonological phrases (chunking of chunks). At the syntactic level, the results of chunking are omnipresent.

What we see as an immediate result of chunking, especially in [S[VO] languages, was already recognized a century ago by structuralist grammarians such as Bloomfield (1933), who called

it “immediate constituents”. Terminals (= elementary elements) are organized into chunks in the chunking network and these chunks are recursively combined into superchunks by the higher-level network, and so on. Anyone who has taken a basic course in syntax is familiar with the outcome of chunking, albeit under the name of *phrase structure analysis*.

Let’s start chunking the following example (1) and keep in mind that (syntactic) processing already starts after the first syllable. The boxes under (1) indicate that the sentence (= final superchunk) consists of two chunks, the second of which is a chunk with two terminal elements.

(1) When complexity is broken down into clarity understanding grows.

The first chunk in (1) consists of several nested chunks with “*into clarity*” as the most deeply embedded one. At the top level, (1) is a superchunk that successfully maps to one of the basic templates of an English declarative, namely (2).

(2) [(XP) [Subject [V_{fin} (YP)]]]

Already at the second item of (1), viz. *complexity*, top-down (= by grammar) assisted chunking can narrow its operation horizon. At this point it becomes clear that ‘*when*’ introduces an embedded wh-clause since it is not followed by a finite auxiliary. Hence, it cannot be a main clause. However, this information can become operative only if chunking – also known as “shallow parsing” in language processing – works simultaneously bottom-up and top-down, that is, input driven and knowledge (= grammar) driven at the same time, accompanied by chunk-internal parsing.

The grammar of a given language determines the chunking templates for this language. Language users have abstracted them in the course of language acquisition. In English, “*When complexity*” does not fit into any template of simple chunks. So, they must belong to two different chunks: [*When [complexity ...]*]. This beginning corresponds to the template of an indirect-question sentence, and it serves as a predictive guide for the ongoing chunking activities on incoming material. At the point when “clarity” has been integrated in the chunk and “understanding” comes in, this indicates that there is a super-chunk, with the already completed chunk as its first chunk. We will not go into details here. This example is presented merely for initial orientation.

At this point, it seems appropriate to clarify some conceptual questions. Why do we need to talk about chunking if we already have an analysis of phrase structures? The simple answer is “cognitive priority”. What we see after having analysed the phrase structure of a given expression is in a causal relation to chunking. Chunking is the cognitive process that maps strings on structures and the properties of chunking are reflected in phrase structures. Therefore, the former provides the explanation of the latter. Phrase structure is grammaticalised chunking. The ways in which complex linguistic expressions are chunked are the ways in which our brain is able to grammatically break them down. In other words, the domain-general cognitive capacities of chunking channel the ways that lead to particular phrase structures. Present day grammars are the present-day versions of cognitive systems with a long⁵ (albeit unknown) history of cognitive evolution.

⁵ Archeology traces homo sapiens back 300.000 years (Richter et al. 2017) and geneticists calculate the split from our last common ancestor with Neanderthals to be between 550.000 and 765,000 years (Stringer 2016).

Cognitive evolution operates in the same way as biological evolution, but in a different domain, on a different substance; see Haider (2021a,b). Evolution is the interplay of random variation plus constant sieving out (‘selection’), with the resulting adaptation to the selecting environment. In the cognitive evolution of grammars, the object of evolution is the cognitive “language app”, with grammar as its main component. The selection environment is the ensemble of our human brain resources we rely on in language processing. One of these resources is the ability to chunk the stream of information. The properties of chunking are therefore reflected in the structures determined by grammars of human languages as a result of cognitive evolution. Grammar variants that impede chunking have no chance to spread. Chunking-friendly variants will prevail in grammar acquisition.

Does this mean that the properties of grammars reduce to properties of parsing? No. This means that the evolutionary development of the grammars of human languages over the last 300 millennia has led to the adaptation of grammatical properties to their selection environment, that is, our neuro-cognitive processing capacities. Just as aquatic mammals have adapted to their environment through biological evolution and look like fish today (cf. whales, dolphins), grammars have adapted to their cognitive environment in cognitive evolution – see Haider (2021a,b) – and the way chunking works is a crucial component of this environment.

Let me add a final comment on the relation between chunking and parsing. Chunking is a domain-general information-processing strategy of our brain.⁶ Parsing is the domain-specific activity in language processing. It consists of “shallow parsing”, also known as “chunking”, and “full parsing”, which tracks all the grammatical details across and within chunks. “It’s all parsing-based” would be a wrong description of grammatical properties, therefore.

Parsing is chunking-based because chunking is the domain-general cognitive resource for the domain-specific activity called (shallow) parsing. Grammars determine the linguistic structures that meet the requirements of chunking in a given language, and grammars do this because they are the result of evolutionary processes in the cognitive evolution of particular cognitive systems in which efficient chunking is part of the selection environment. Grammar variants that define chunking-unfriendly structures have little chance of spreading. They are sieved out in language acquisition and language use. The variants that are winners in the course of cognitive evolution are those that are chunking-friendly. Such an evolutionary explanation has no need for a hypothesis of an innate UG. The properties listed at the end of the following section under (3) are some of the chunking-friendly properties of syntactic structures that can be observed cross-linguistically.

2. The E-C-S model of chunking – How it is reflected in grammars

The ECS model describes the core architecture of chunking, as outlined in Figure 1. Elementary items are grouped in chunks by the chunking network, whose output is input to the super-chunking network. In the different grammars of languages, virtuoso forms of chunking can be recognised, depending on which properties are used to link the elements of a chunk. Let’s talk

⁶ “Mathematically, the ‘chunking principle’ can be viewed as the transformation of a chain of metastable states along a transient process to the chain of groups of such states. It is a key dynamical idea that nature may use to make cognitive information processing more effective in the context of a complex environment.” Rabinovich et al. (2014: 3).

about *syntactic* chunking, because it is easy to illustrate. Basically, there are two ways of identifying elements of the same (super-)chunk, the morphological and the topological way. Latin is well-known for its morphological identification while English identifies chunk members topologically. (1a) and (1b) are translationally equivalent but syntactically different.

(1) a. Cūriō Mārcium Uticam nāvibus praemittit. (Caesar, Bell. Civ. 2.24.1)

Curio_{Nom} Marcius_{Acc} Utica_{Acc} ships_{Abl} ahead-sends

b. Curio sends Marcius ahead with the ships to Utica'

In (1a), for instance, Cūriō and Mārcium could swap their place *salva veritate* and *salva grammaticalitate*. In (1b), however, this would result in a different message. What morphological case does in (1a), prepositions do in (1b), namely identify the core arguments. As for chunking, (1a) is a chunk consisting of five terminal elementary items. (1b), however is a fairly complex superchunk, consisting of at least three sub-chunks, namely the two PPs and the VP.

Latin and English illustrate a trade-off relation between chunking economy and morphological coding economy: no (nominal) morphology, but many chunks in English; rich morphological coding, fewer sub-chunks in Latin. Chunking is a task subserved by the procedural-memory network, while morphological decoding and encoding is subserved by the declarative-memory network. This is the key for understanding the diachronic dynamics (Haider 2021a,b).

In other words, topological coding, i.e., coding by the respective positions in an articulated phrase structure (as in English) is at the expense of chunking (procedural memory). Morphological coding, i.e., coding by the respective form paradigms (as in Latin) is at the expense of the declarative network. What is more economical in terms of the brain's processing load? Anyone who was confronted with the two languages for the first time at school knows this from personal experience. Declarative coding in inflectional languages with their varieties of paradigms is more resource intensive. Here is an example (2). In Latin, the phrase “*in this so troublesome a duty*” is discontinuous. Its three parts are identifiable though by agreement (number, gender), but they are not 1:1 in form and function. Inflectional coding of the same value may formally differ along numerous inflection classes. A topologically encoding grammar like that of English in the translation in (2) seems to be simpler and more efficient, especially since it also involves fewer positional variations.

(2) Ac ne *in hoc* quidem *tam molesto* tacebant *officio*.

and not *in this*_{Agr} even *so unpleasant*_{Agr} remained-silent *duty*_{Agr}

‘And not even *in this so troublesome a duty* were they silent.’

[Petronius Arbiter, Cena Trimalchionis §31. Taken from Bolkestein (2001: 255)]

The diachronic grammatical changes in the Romance and Germanic language families have all taken the very same direction. The Latin-like morphological coding systems eroded and ultimately disappeared when topological coding became established, both on the way to modern English, to the contemporary continental Scandinavian languages, as well as in Romance languages. On the other side, there is no known language that would have replaced topological coding with a fully-fledged morphological system. Topological coding is a one-way road to a point of no return (Haider 2023). Material disappears that cannot be regained again.

Topological coding not only relieves the declarative system, it also facilitates chunking since topological coding reduces constituent order variation. In vernacular Latin, and in similar languages, the five items of the single chunk (1a) can be arranged in 5! ways (5! = 120). In English, the subject only precedes the verb and the direct objects only follows the verb immediately. Thus, chunks and sub-chunks are predictable with a higher chance of accuracy in English than in languages with variable constituent orders.

Once chunks are categorised, they can be identified and told apart. Grammars typically rely on *endocentric* categorisation.⁷ The category of the chunk is the category of an element that is associated with any other chunk within the chunk. Endocentric categorisation is monotonic. The category does not change with the size of the chunk. Such elements are familiar to us as lexical heads of phrases.

Chunking is the key to understanding many crucial features of phrase structures in natural languages, from universal structural asymmetries to the positioning patterns of heads and their dependents. At this point, it is important to realise that chunking is a time-pressured activity. In grammar theory, phrase structures are just as timeless as maps are. But as soon as one uses a map to plan a journey, time suddenly becomes important, especially when it comes to finding the fastest route. The “here & now-bottleneck” favours the fast tracks. What properties of phrases make chunking fast and efficient? This is the crucial question and the answer will tell us a lot about phrase structures. Their properties are properties that result from automatic dynamic chunking and (3) lists some of them.

- (3) a. Binary branching
- b. Right-branching (the earlier, the higher – the later, the deeper)
- c. Fillers precede their gaps (as a corollary of right-branching).
- d. V-Clustering with head-final VPs (avoidance of centre-embedding)
- e. Left-Left adjunct constraint
- f. Optimal chunk size

2.1 Binary branching

There is ample cross-linguistic evidence for “1-constituent only” restrictions as immediate evidence for binary branching. Here are a few examples. In V2-languages, a *single* constituent precedes the finite verb in declarative main clauses. In languages with determiner systems, a *single* ‘genitive phrase’ precedes the noun phrase.⁸ There are languages that front a *single* wh-phrase. In each of these cases the property involves the notion of a single *constituent*. Therefore, the obvious question is what a constituent is. This is determined by the grammar of the given language and grammars also show us in their results – directly or indirectly – how constituents are structured. Binary branching is a property of phrases that guarantees that each expression ends up as composed of minimal and therefore structurally predictable chunks.⁹ Consequently,

⁷ Principally, phrases can be identified also in a non-endocentric way, as Kayardild shows, a moribund Tangkic language (South Wellesley Islands, Australia), with “complementized” clauses: “*For example, when the OBLique case is used in complementizing function it appears on all words of the complementized clause.*” Evans (1995: 101). In other words, *each one* of the (potentially scattered) terminal elements of an embedded clause is *morphologically identically* marked as being clause mates.

⁸ Trump's destruction of the democracy – *Trump's the democracy's destruction

⁹ In technical terms, it has a 1-1 value-address format. There is exactly one value and there is exactly one node on the projection line that is its structural address.

it can be divided into, and composed out of, such chunks, which is the essential property. Here is an example, with minimal chunks in (4a) in comparison with chunks that would result from ‘flat’, that is, multiply branching structures (4b).

- (4)a. [wenn [der Wind [den Bäumen [die weiße Pracht [von den Ästen weht]]]]] German
 when the wind the trees_{Dat} the white splendor_{Akk} off the branches blows.
 ‘when the wind blows the white splendor from the branches of the trees.’
 b. [wenn [der Wind [[den Bäumen] [die weiße Pracht] [von den Ästen] weht]]]
 c. [[Von den Ästen geweht] [hat [der Wind den Bäumen die weiße Pracht - -]]]
 off the branches blown has the wind the trees_{Dat} the white splendor_{Akk}
 d.*[[Den Bäumen die weiße Pracht] hat der Wind von den Ästen geweht
 the trees_{Dat} the white splendor_{Akk} has the wind off the branches blown

In (4c), the clause-initial portion is a single phrase and thereby a single chunk. This is congruent with a constituent in (4a), but not in a flat structure as in (4b). Here, it would be part of a chunk that would not correspond to a *single* constituent. It is expected to be as deviant as (4d), which is another version with two sub-chunks of (4b). “*Von den Ästen geweht*” is a constituent in (4a) whereas “*den Bäumen die weiße Pracht*” is not.

Why would a grammar that structures a sentence as in (4a) prevail in cognitive evolution over a grammar that assigns the structure (4b)? Firstly, (4a) guarantees a rigid format of the chunk-superchunk relation. Each superchunk consist of at most two chunks. On the other hand, the grammar of (4b), according to which the VP would consist of a *set of chunks* (in a sister relation), does not have a deterministic chunking structure. There is a number of sub-chunks, each of which could be a superchunk but to determine the exact number, one would have to see into the future (which is not a natural human gift).

Binary-branching does not determine the linear order; it merely determines the hierarchies of chunking. *Directionality* is an essential property in language processing, however. Linguistic expressions are ordered linearly. This is merely another way of describing the one-dimensional structure of utterances on the phonetic-phonological level, which are determined by the order relations “before” and “after” of the terminal elements. The chunking hierarchies would remain identical under mirror-transformation. (5) would have the same chunking hierarchy as (4a), but sentences structured like (5) do not exist. What would be wrong with them?

- (5) [[[[[Weht von den Ästen] die weiße Pracht] den Bäumen] der Wind] wenn.]
 blows off the branches the white splendor the trees the wind if

In (4a), the utterance starts at the highest position of the structure. In the mirror-image world, the utterance starts at the lowest position (5). It should come as a surprise that in the Generative theory of merge & move, this structure would be predicted to be perfect. Merger would start with the verbal head as the nucleus and continue to merge elements on the right (viz. on the time line of uttering), in the course of completing the utterance. Once it has reached the top, the job is done. Alas, language processing does not work in this way.

If it had to work like this, the task would be a hurdle course. If chunking had to begin with the lowest position, it would be entirely unclear how deeply embedded this position is. Therefore, the parser would have to constantly backtrack and change the structure. In (4a), however, parsing is from top to bottom and not from bottom to top, as in (5). The beginning is the highest

position in the sentence structure and therefore remains constant. This is guaranteed by a right-branching sentence architecture (see the following sub-section). In addition, top-down information (from grammar) and bottom-up information (from incoming expressions) interact effectively in this way.

2.2 Right-branching – The later, the deeper

In Haider (1992), (1995), (1997) and subsequent work – see the survey in Haider (2013: ix-x) – empirical and theoretical reasons have been presented for a universal asymmetry of phrases with lexical and lexical-functional heads.¹⁰ Universally, they are *right-branching*. The node on the projection line of a lexical head always *follows* its sister constituent that represents an argument of the head (6a). It is the node on the *right* side. Left-branching projections (6b) do not occur. In (6), h° stands for a lexical head. Let us assume its argument structure provides three arguments, which are ranked $A < B < C$. In this case, the binary branching structure is (6a). The left-branching structure (6b) is not in use in grammars of natural languages.

- (6) a. $[_{HP} A [_{H'} B [_{H'} C h^\circ]]]$ *right-branching* (A,B,C as arguments of h°)
 b. $*[_{HP} [_{H'} [_{H'} h^\circ C] B] A]$ *left-branching* (A,B,C as arguments of h°)

A grammatical principle responsible for the exclusion of (6b) has been presented first in Haider (1992), namely the BBC (Basic Branching Constraint). Why should grammars abide by this principle? The answer is this. Cognitive evolution of grammars selects grammars that impose the phrase structure (6a) rather than (6b) because of the selection effect of chunking in the evolution of grammars. Right-branching structures are chunking-friendly, left-branching structures are chunking-unfriendly. They are chunking unfriendly because utterances would have to start at the deepest position in (6b). The unknown depth of this position¹¹ could be determined only at the very end of the whole expression (see the brackets on the left-hand side). In (6a), the structure starts at a well-known point that remains constant, namely the top node. In (6a), chunking can proceed from outside to inside. This is how shallow parsing proceeds. Internal details of the chunks have to wait for ‘later’. In (6b), it would have to start from the inside and reach the outside only at the end. Ironically, in the Minimalist Program, (6b) is expected to be the universal structure since *merging* starts at the head-position and proceeds from there on.

There are only a few languages whose history we can trace back for a millennium or more and they teach us that there is a type of phrase structures that came into being much later than structure variants of (6a), namely *head-initial* structures (7c), with a preverbal subject position; see the VPs of Germanic VO languages (Haider 2014) or of modern Romance languages. They are a late development since in these languages, *recursive chunking* is at work *within* a verb phrase.

- (7) a. $[X [_{V'} v^\circ Y]]$ basic *head-initial* projection scheme
 b. $[B [_{V'} v^\circ C]]$ lowest projection (of a verb with two objects)
 c. $[A [V^\circ [B [_{V'} v^\circ C]]]$ *recursive* projection (of a verb with three arguments)

¹⁰ It should be noted that the entire merger business of MP principally operates without directional restrictions and without time considerations. In other words, the left-branching mirror image of English would be a structurally well-formed language, at least for the MP.

¹¹ In other words: the number of left brackets to be opened

The recursion of the basic scheme (7a), with (7b) in place of “Y” of (7a) yields (7c), the by-now well-known VP-shell structure of complex head-initial VPs. Since (7c) is the (recursive) projection of a *single* verbal head V° , the lower V° in (7c) is empty. The grammatical relation between the lower and the higher V° is familiar from head-chains. The lower one is a ‘trace’ of the higher one. This has been presented first in Haider (1992), and already there, it is pointed out that the same kind of shell-structure occurs also in complex head-initial NPs, as illustrated by (8), with the genitive DP c-commanding and binding the reflexive.

- (8) [**Fotos** [des Mannesⁱ_{Gen} [-- von sichⁱ]]]
 photos (of) the man [-- of himself]

The shell structure of the head-initial sentences is the forced result of the interaction of two antagonistic requirements. In a *right-branching* structure (= requirement 1), a chunk-initial *head* of a phrase with more than one argument, cannot be in the foot-position of the phrase (= requirement 2 = endocentricity). On the other hand, the early presentation of the head is an advantage for chunking and fine-tuned parsing since the head is the address of the lexical argument structure. The grammatical means for early heads of phrases despite endocentricity is *recursion within the projection*, in complex VPs as well as in complex NPs.

Diachronically, these structures are a late development in the evolution of grammars. First there are flexible-order languages. The [S[VO]] structure is not at the beginning of the evolution of sentence structures (as Generative Grammar assumes), but it is a fairly recent result. No language is known to have developed from a typical [S[VO]] language with little to no grammatical morphology into a language like Latin, Sanskrit, or Old Persian with a morphologically rich case and agreement system. The other direction, however, is attested beyond reasonable doubt (see the history of Germanic and Romance VO languages). No other language family than the Indo-European one is available and researched in the same historical depth and breadth.

2.3 Fillers precede their gaps

In filler-gap constructions, the filler precedes the gap because the filler is in a displaced position and this is higher up in the structure. In *right-branching* projections, higher up is to the left. Let's go back to the *left-branching* mirror-image world (6b) for a moment and imagine how filler-gap relations would look like. The gap would precede the filler. In local as well as non-local displacements, the gap would be in a lower, i.e. preceding, chunk, with the filler coming towards the end of the utterance. In other words, a gap would have to be identified first, even before it is known that a filler is present and that a gap occurs. In the properly branched, real existing grammatical world, the presence of a filler is the trigger for the search of a corresponding gap.

Examples of typical filler-gap construction are (local or non-local) wh-fronting of interrogatives, pronoun-fronting in relative clauses, or verb-second declaratives in V2-languages, with a single constituent fronted to the clause-initial position. None of these constructions has a mirror-image equivalent in other languages. In other words, there are V2 languages but there are no V-penultimate languages, and there are no languages with relative clauses that precede the noun and obligatorily put their displaced relative pronoun into the final position in a relative clause. Clause final wh-items occur, but they are not placed into the final *top*-region of a clause. They are clause-final if in the given language, the focus position is in the lowest position and therefore

clause-final. There is no counterpart of English or Slavic languages, with *wh*-phrases displaced into a right-hand top position in a clause.

2.4 V-Clustering in clauses with head-final VPs

Head-final VPs are structurally much simpler organized than head-initial ones. Their arguments are lined up along the right-branching projection line with the verbal head in the foot position, as sketched in (6a). This phrasal architecture is perfect for simple verb phrases but becomes a challenge for chunking as soon as there are several nested verb projections. The problem is known as centre embedding. Let's compare a head-final version (9a) with the head-initial version (9b). In (9a), the stacked VPs would start with the most deeply embedded Phrase, namely “das Ende”. The chunking parser cannot determine the precise depth of embedding. In (9b), on the other hand, the VP starts with the head of the top VP. This chunk can be precisely identified and its internal substructure unfolds head-by-head. The evolution of OV grammars has provided a way out of this difficult constellation. The solution is *verb clustering* (VC), as in (9c).

- (9) a. [_{VP} [_{VP} [_{VP} das Ende der VP erraten] können] sollen]
 the end of the VP (to) guess be-able shall_{Inf}
 b. [_{VP} shall [_{VP} be [_{VP} able [to [_{VP} guess the end of the VP]]]]]
 c. [_{VP} das Ende der VP [_{VC} [erraten können] sollen]]

Clustering reduces the region of uncertainty to a *local region* at the end of the verb phrase. Instead of nested VPs, i.e. one per (quasi-)auxiliary), there is a verbal cluster, with a single VP with clustered verbs. This chunking-friendly grammatical solution invites a clustering alternative (10b) also for centre-embedded infinitival clauses (10a). The existence of this alternative becomes directly recognisable for instance when passive is applied to the cluster (10c), see Haider (2010: 319, 2024). In the passive of (10a), the clausal construction, on the other hand, the embedded object is not affected (10d).

- (10) a. dass jeder [PRO [PRO dies_{en} Kollegen zu informieren] zu versuchen] vergessen hat
 that everyone [[this colleague_{Acc.} to inform] to try] forgotten has
 ‘that everyone forgot to try to inform this colleague’
 b. dass jeder dies_{en} Kollegen [[[zu informieren] zu versuchen] vergessen] hat]_{VC}
 c. dass dies_{er} Kollege_{Nom.} [zu informieren zu versuchen vergessen wurde]_{VC}
 that this colleague to inform to try forgotten was
 d. dass [PRO [PRO dies_{en} Kollegen _{Acc.} zu informieren] zu versuchen] vergessen wurde
 that [[this colleague_{Acc.} to inform] to try] forgotten was

For a detailed study of these infinitival clustering constructions in comparison to recursive clausal embedding, please consult chapter 7 of Haider (2010). The important point for the present discussion is this. The clustering option is an OV phenomenon. In [S[VO]] languages, clustering as in (10b) and therefore also passivisation of the cluster, as in (10c), are unavailable. The reason is easy to understand. In a language with head-initial structure, centre embedding does not occur in the complementation architecture; see the second gloss in (10a).

From the theoretical point of view of a grammar, there is nothing wrong with (10a). It is the head-final version of the very same infinitival complementation that is found in head-initial structures in languages like English. The Minimalist Program, with its purely algebraic view, is actually confronted with two problems. First, there is no reason to give up (10a) for (10b). It

is a perfectly merged and fully grammatical specimen. Second, the MP has no explanation for the fact that clustering in infinitival complementation is restricted to head-final contexts. Chunking makes the difference. Clustering avoids the jam of centre-embedded structures at the “Here & Now bottleneck” of the 1D-to-2D mapping¹² of phonological strings to syntactic structures. This jam does not occur with head initial structures.

2.5 Left-Left-Adjuncts

Head-initial structures also pose a problem for chunking and the evolution of grammars has found a way to solve it. The grammatical constraint is the *Left-Left constraint* (LLC), described in Haider (2022). The head of the *left* adjunct of a strictly¹³ *left-headed* (=head-initial) phrase must be adjacent to the host phrase of the adjunct.

(11) [... [Adjunct ... A° (*...)] [XP ... [X° ...]]]

In English, VPs (12a) and NPs (12b) are head initial and the LLC regulates both. In German, NPs are head initial, VPs are head final. So, the LLC applies to prenominal attributes (12c) but not to preverbal adverbials (12d).

- (12) a. She has more carefully (**than anyone else*) [*analysed* the problem]
 b. a much more careful (**than this one*) *analysis*
 c. eine sehr viel sorgfältigere (**als andere vorher*) *Analyse* des Problems
 a much more careful (than other-ones before) analysis (of) the problem
 d. Sie hat alles viel sorgfältiger (*als jeder andere vor ihr*) *analysiert*
 she has everything more-carefully (than anyone else before her) analysed

Why would grammars impose such a restriction? The source of the problem is the *addressing* (aka linking) of non-arguments within phrasal chunks. The decisive restriction is not the prohibition of centre-embedding of extensive adjuncts. The LLC does not correlate with size, neither in VO like English, nor in OV like German. German APs are head-final. Hence their heads are always adjacent to the NP. Therefore, they trivially fulfil the LLC, irrespective of the size of the adjuncts, as in (13), as long as the head is the final element.

- (13) eine [mit solchen Phasen, die beliebig komplex sein mögen, *einhergehende*] Erschwernis
 a [with such phases, which arbitrarily complex be may, *associated*] aggravation

Why would the head of the adjunct have to be in the final position, even if the adjunct is a head-initial constituent, as in (14)?

- (14) a. [[extremely *happy*_{A°} (*with the result)]_{AP} teams]_{NP}
 b. [[sufficiently *familiar*_{A°} (*with it)]_{AP} experts]_{NP}

The crucial property seems to be this. A left adjunct of a left-headed phrase is in no-man's-land when it comes to linking it. Head-initial phrases link their dependents to the *right*, that is, the depending phrases have to follow. If a phrase precedes such a head, it is not within the linking domain. In grammatical terms, the head of the left adjunct of left-headed phrase cannot be

¹² Phonological arrays are one-dimensional (= 1D). Syntactical and semantical structures are hierarchically organized box-in-box structures, that is, 2-dimensional (= 2D) objects.

¹³ In T3 languages, i.e. languages with flexible head positioning, it is (predicted to be) absent with adjuncts left-adjoined to a phrase whose head happens to be phrase-initial; see Haider & Szucsich (2022a,b).

linked by the head of the host phrase (because of the directionality mismatch). This is the grammatical side. The adjunct must therefore connect itself to the phrase (see Haider 2022: 203-204), indicated by “ $\oplus$ ” in (15).

(15) $[XP [Adjunct \dots A^\circ] \oplus [XP \dots [X^\circ \dots]]]$

The adjunct phrase must *import* its structural address “[XP” in (15) since it does not receive one from X° or its projections because it is not within the directionality domain of X° . This happens via the head of the adjunct in a strictly local relation to the host phrase. The effect is a phrase-final head that marks the end of the adjunct and the beginning of the phrase it is adjoined to. Why would this be crucial only for head-*initial* projections? In terms of chunking and parsing, the LLC as a grammatical restriction seems to be a reflection of the beneficial punctuation effect of directional head positions in phrases in the evolution of grammatical structures.

2.6 Optimal chunk size

Linguists are used to refer to verbs as predicates, and to subject and objects as arguments of these predicates. Logicians point out to us that logically, the number of arguments a predicate can have is unbounded: “*In natural language ternary predicates are enough to express the most complex verb pattern you can get, but logical languages can handle any number of arguments.*” Van Benthem et. al. (2016: 85).

Even if we admit (16) as a rare exception¹⁴ on the one hand and accept time and place adverbs as logical arguments on the other hand, the fingers of one hand are still enough for counting the arguments of a verb in everyday life. One may ask therefore what prevents grammars from offering n-ary predicates, with $n > 4$.

(16) She₍₁₎ bets him₍₂₎ ten dollars₍₃₎ [that it won’t rain]₍₄₎

Generative grammar would presumably fall back on UG (Universal Grammar). Since UG is considered innate, the property would be derived from an innate property. After all, the fact that we have (only) five fingers is innate too. Nice try, but to no avail. Tabin (1992) reviews the genetic basis and the role of hox genes for limb development. However, a genetic basis for the number of arguments of lexical heads has not yet been found and it is not worth waiting for it. The odds are close to certainty that there is none.

The obvious place to search for an explanation is chunking, in particular in relation to the here & now bottleneck of our short-term memory capacities for processing. In reception as well as in production, the relations between the heads and its arguments with concomitant case & agreement checks in many languages strain the capacity of the short-term memory at the cognitive work bench.

David Dirlam, a mathematician, calculated the most efficient chunk size: “*A mathematical analysis of hierarchical organizations suggests that the most efficient hierarchy for variable order searches has either 4 or 3 items per chunk.*” Dirlam (1972: 355). He asked what chunk size would *minimize* the expected number of total node and item accesses. At each level of the hierarchy, the *minimum* would occur with an average chunk size of 3.59 items.

¹⁴ The German version is a verb with two PPs: *Sie wettet mit ihm um 10 Dollar, dass es nicht regnen wird.*
she bets with him for 10 dollars, that it not rain will

In the past fifty years, since Simon (1974), there is a “*remarkable degree of similarity in the capacity limit in working memory observed with a wide range of procedures.*” Cowan (2000: 114). In his target paper on the same question, Cowan (2000: 108), reviews the issue from the experimental side. “*The empirically observed capacity limit of about four chunks corresponds to what has been predicted for how many items can be advantageously processed in an ungrouped manner when the search is exhaustive (MacGregor 1987).*”

“*The magical number 4 in short-term memory*” is a relevant part of the selection environment of grammars. Variants that enhance the efficiency of identifying the core arguments of a verb or noun prevail in the cognitive evolution of grammars. There is no grammar-internal restriction, neither in terms of morphological¹⁵ nor topological implementation. A linguistic observer merely realises that there are no verbs with an argument structure with more than 3-4 core argument positions, which corresponds perfectly to Dirlam’s 3,59 items per chunk. It should be noted that these chunks are not terminal elements, but rather the (complex) elements that make up the VP as a (super) chunk.

The size of the argument structure is a perfect example of the interplay of cognitive and grammatical factors in the cognitive evolution of grammars. Linguists can arrive at correct generalisations about the *form* of argument structures without recourse to cognition, as long as they refrain from asking what the generalisations result from. If they ask *why*, they should know where to seek the answer in the *conditio humana*, namely in the *cognitio humana* as the selecting environment in the evolution of grammars. Grammars restrict the argument structure but not *in order to* make chunking easier and more efficient. They restrict it because these variants of grammars have succeeded in the sieving-out processes of the cognitive evolution of grammars. This process is going on as we can see if we compare the many different systems of grammars in use. There is no teleology at work, as functionalists would claim. What appears to be teleological is the effect of a constant but blind selection of variants in the (ongoing) course of the cognitive evolution of grammars.

3. Discussion

Chunking operates on serialised input. Linearisations convey information for the forming of chunks and the identification of syntactic structures with the contribution of top-down information by the grammar. In today’s Generative Grammar, serialisation is regarded as a secondary and derivable property. Linearisation matters only at the PF (‘phonetic form’) interface. It is deemed to be the result of a serialisation rule that translates asymmetric c-command relations (recte: asymmetric relations over sets) into linear orders: If, and only if, α asymmetrically c-commands β , α precedes β in linear order.

This instruction runs into a problem at the bottom of structures, when merger starts joining two elements. This is the “*ignition problem*” or “*bottom-pair*” problem for the linearisation operation. If α is merged with β , the structure $\{\alpha, \beta\}$ is symmetrical. It is identical with $\{\beta, \alpha\}$ and the two items c-command each other. Serialisation is not defined in this case. Details aside, the relevant, merger-generated structure of (17a) is (17b). What makes ‘*precede*’ precede ‘*objects*’?

¹⁵ See Finnish and its 15 cases: four for core arguments, six for alternative locative relations, three for comitative-like relations.

- (17) a. Subjects precede objects.
 b. {Subjects {precede objects}}

In a recent paper on structuring and linearisation, Moro & Roberts (2024), henceforth M&R, let us in on the current state of affairs. They assume a boot-strapping repair method for the MP's “bottom-pair” dilemma. One of the two items of any symmetric pair is lifted, either by “roll-up” (18a) or by head-lifting (18b), thereby creating an asymmetric configuration. The “repair hypothesis” is seen as an integral part of the merger-and-spell-out theory.

- (18) a. $\{V^\circ \text{ Obj}\} \Rightarrow \{\text{Obj} \{V^\circ \text{ Obj}\}\}$ OV-base scheme by roll up (M&R 2024, ex. 8)
 b. $\{V^\circ \text{ Obj}\} \Rightarrow \{V^\circ \{\text{V}^\circ \text{ Obj}\}\}$ VO-base scheme by lifting (M&R 2024: ex. 9)

In English (17b), which is representative for head-initial languages, head-lifting (18b) produces (19) which is spelled out as (17a). In German (20a), a language which is head-final for VPs, roll-up of the object (18a) turns the symmetric product of merger (20b) into (20c), which is spelt out as (20a).

- (19) {Subjects {precede {~~precede~~ objects}}}

- (20) a. dass Subjekte Objekten vorangehen German
 that subjects_{Nom} objects_{Dat} precede
 b. {vorangehen Objekten}
 c. {Objekten {vorangehen ~~Objekten~~}}

If M&R were right, the MP would produce chunking-friendly input, at first glance. There would be a simple correlation between linear order and structural hierarchy. Universally, what precedes is higher in structure than what follows. Conversely, what follows is structurally lower than the preceding elements. So far, so simple, but grammars are not quite so simple. Let us move on from such overly simple cases as (17a) and (20a) to more testing cases such as the linearisation patterns of auxiliaries and quasi-auxiliaries in clauses of V-final languages:

- (21) dass darüber *diskutiert werden müssen wird* German
 that on-this *discussed be must_{Inf} will*
 ,that this will have to be discussed‘

Merger would work as in any other language, e.g. English, and would result in the following structure (22a), which, taken at face value, would have to be spelled out as (22b), however. So, (22a) cannot be the structure relevant for the linearisation in (21).

- (22) a. {dass {{{darüber {diskutiert ~~darüber~~}} werden} müssen} wird}}
 b. dass wird müssen werden diskutiert darüber (*German)
 that will have-to be discussed it-on

Evidently, if we calculate back in the linearisation procedure, the structure of (21) cannot be nested as in (22). In the given serialisation (21), “*wird*” (will) is the *final* element in the clause. So, it must be the most deeply embedded one. If it were higher in the structure, as in (22a), it would have to precede, as in (22b).

Should we really be surprised that the MP theory has no *straightforward* way to achieve the ordering of the auxiliaries known from the OV languages? If we followed the logic of roll-up and lifting, the structures of OV languages would require very complex repair logistics before

spell-out. The entire architecture created by merger has to be turned upside down completely. Every VP in (24a) would have to be rolled up. I spare the readers the impressive number of nested brackets needed for ending up with “*darüber*” and “*diskutiert*” asymmetrically c-commanding “*werden*”, and so on. The confusing details can be found in the appendix of this paper.

It should be noted that this mess is merely the result of inadequate theory. Compared to the “down-to-earth” architecture of OV presented in Section 2.2, the roll-up game at least provokes the question of why grammars should bother to do that, only to end up as OV. Those who believe that OV languages are so bizarrely organised may keep on dreaming about it, but it is possible to wake up for the better.

The prematurely praised chunking-friendliness of MP structures is completely lost as soon as MP is confronted with the reality of languages that are not VO. The preceding reasonings deal with a purely theory-internal issue, disconnected from the empirical reality. The MP attempts to work without assuming a vector from which the serialisation direction follows directly. Linguistic facts tell a different story.

There are languages with different head positioning depending on the *category* of the head item. In German, the VP is head-final (23a), the NP is head-initial (23b). There are even category internal differences. Most adpositions are head-initial (23c) (prepositions), for instance, but some are head-final (postpositions), see (23d).¹⁶

- (23) a. ein Problem [$\leftarrow_{V^0}$ lösen]
 a problem solve
 b. das [Lösen] $_{N^0 \rightarrow}$ eines Problems
 the solve_{Inf} a problem_{Gen}
 c. [laut] $_{P^0 \rightarrow}$ der Washington Post
 according-to the_{Gen} Washington Post
 d. der Washington Post [$\leftarrow_{P^0}$ zufolge]
 the_{Gen} Washington Post according-to

The fact that serialization is a property that covaries with lexical categories is a strong indication that it is associated with the lexical category of heads. Sometimes it is even restricted to a single lexical item.¹⁷

Evidently, what we see here is a vector property of heads of phrases. The bottom-pair problem vanishes into thin air once one is ready to realise that at the bottom, one of the two elements must be a head (see: endocentricity). This element provides the vector information, that is, the *linking directionality* of the head element (of this category) and thereby the order of head and complement (Haider 2015). This also solves the VO-OV problem in one go. The *branching directionality* is another empirical issue (see Section 2.2). It cannot even be adequately discussed within MP theory, since directionality has no place in the theory, except as a secondary effect of spell-out at the PF interface.

¹⁶ entlang – along, gegenüber – against, halber – sake, wegen – because of, zufolge – according to.

¹⁷ In *all* Germanic languages there is an item-based outlier: *enough* and its cognates follow, while all other degree modifiers precede: rich *enough* – sufficiently rich. This is evidently a lexically based serialisation property.

The preceding discussion is intended to illustrate the degree of disorientation that arises when empirical questions are treated from a purely axiomatic standpoint that is completely detached from the cognitively determined empirical conditions of grammars. One feels sorry for the authors when they summarise at the end: “*We are aware that these typological remarks require a great deal of further substantiation [...] but we believe that this alternative view is of interest also because it brings cognitively relevant elements into the explanation.*” In the paper, there is not a single reference to a property of cognition or a ‘cognitive element’. It contains three tokens of the word “cognitive”, two of which are part of the preceding full quote, and one is about language acquisition. However, the paper is an impressive exercise in theoretical reasoning far removed from linguistic reality. Instead, any thought of chunking would be time well spent.

As diachronic research shows, the path to today's [S[VO]] systems – see the grammatical history of English or of continental Germanic languages – leads from languages with variable ordering of constituents (viz. T3-languages) to languages with a fixed head position and subsequent reduction of morphological coding, especially in [S[VO]] languages; see Haider (2014). The linguistic processing load is gradually shifted from the general declarative-memory system to the fast and automatised special-purpose procedural system, with chunking efficiency as one of the predominant selection factors in the cognitive evolution of grammar systems (Haider 2023). The MP is completely out of its depth with such facts and problems. It fails to provide sensible solutions but seduces its followers into irrelevant sophistry instead of leading them to successful theory building with hypotheses that can be tested empirically.

4. Concluding remarks

Grammar research from the perspective of human chunking abilities and (neuro-)cognitive chunking procedures leads directly to meaningful and empirically adequate answers to central questions of grammatical structuring such as:

- Why are syntactic structures right-branching?
- Why are syntactic structures binarily branching?
- Why is verb clustering a property of head-final VPs?
- Why do verbs have not more than four core arguments?

The answer to these questions arises directly from the way how chunking works. Chunking plus directional grammatical linking is sufficient for deriving a wide range of grammatical properties (Haider 2015). These are the basic templates:

- (24) a. The basic *structure* template: $[H^*... [H^*... [H^*...]]]$ (vs. $*[H^*[H^*[H^*...]] ...]$)
 b. The head-*final* template: $[H^* XP \Leftarrow [YP \Leftarrow H^\circ]]$ (“ H^* ” = *projection* of H°)
 c. The head-*initial*-template: $[H^* H^\circ \Rightarrow [XP [H^\circ \Rightarrow YP]]]$ (plus recursion; see (7c))

The head-final template (24b) is congruent with the basic-chunking template (24a), the head-initial template (iii) is congruent, too, but the linking directionality is inverse. Therefore, recursion is necessary for instantiating the canonical linking of XP in (24c) by H° , whence the ‘shell’ structure of complex head-initial phases. No innate UG is needed for explaining this and the other properties (i.e. binary branching, left-bound filler-gap structures, V-clustering in OV, size of the argument structure of a lexical head). It is the outcome of the cognitive evolution of grammars (Haider 2021a,b), with chunking efficiency as a strong selection factor.

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Appendix – a brainteaser.

Target linearisation:

- *dass darüber diskutiert werden müssen wird*¹⁸ (= ex. 21)

Base structure after merger (Minimalist Program):

- [dass {wird {müssen {werden {diskutiert darüber}}}}}]

Linearisation of the base structure by asymmetric c-command:

- *dass wird müssen werden diskutiert darüber

The structure that is required for the target linearisation:

- dass {darüber {diskutiert {werden {müssen {~~müssen~~ wird}}}}}

Brainteaser: How to arrive at the above c-command structure, starting from base structure?

¹⁸ Please note that this order pattern is no peculiarity of German. “In languages with dominant order SOV, an inflected auxiliary always follows the main verb” (Greenberg 1963:67). “In OV languages, auxiliary verbs normally follow the main verb, while in VO languages they normally precede.” Dryer (2007: 90).

Difficulty of the OV-type structure: last in (by merger) = last out (by spell out).

Problem 1: Roll-up cannot be applied before the *last* auxiliary is merged since roll-up has to target the *relatively* highest position (below C°).

Problem 2: When merger is completed, the entire base structure (below C°) would have to be rolled-up in a *nested* way so that the desired structure emerges:

- [dass {wird {müssen {werden {diskutiert darüber}}}}}]
- [dass {diskutiert {wird {müssen {werden {diskutiert darüber}}}}}]
- [dass {diskutiert {werden {wird {müssen {werden {diskutiert darüber}}}}}}}]
- [dass {diskutiert {werden {müssen {wird {müssen {werden {diskutiert darüber}}}}}}}]
- [dass {darüber {diskutiert {werden {müssen {wird {müssen {werden {diskutiert darüber}}}}}}}}}]

This is a 5-step derivation that a simple 6-word sentence demands of speakers of German in a roll-up theory. Let's note that this involves two highly questionable assumptions (delayed reordering, nested reordering). It is fairly safe to assume that such grammars will not materialise ever. Here is the competing structure assignment, with a directionality vector, and without any need for reordering at all.

- [CP dass [VP darüber ←[[[VC diskutiert ←werden] ←müssen] ←wird]]]